

Sameer Khan

SENIOR MACHINE LEARNING ENGINEER | DOCUMENT AI
& MULTIMODAL AI | LLM/RAG | AWS MLOps

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Document AI & Machine Learning Engineering professional targeting Senior/Lead Machine Learning Engineering roles focused on designing and scaling production-grade AI/ML systems, leveraging extensive experience in complex enterprise AI solutions to drive robust model architectures, scalable ML pipelines, efficient deployment, and high-performance production platforms. Searching for opportunities to contribute as a strong technical individual contributor with an architecture-oriented approach to solving large-scale ML challenges.

PROFILE SUMMARY

- Senior Machine Learning Engineer with over 9 years of experience in building and productionizing AI/ML solutions across Document AI, Multimodal Deep Learning, Computer Vision, NLP, Generative AI, LLM/RAG, and MLOps, with end-to-end exposure spanning data processing, model development, evaluation, optimization, deployment, and production monitoring.
- Document Intelligence specialist experienced in document classification, OCR/OCR-free processing, information extraction, NER, signature verification, and document deduplication, contributing to high-scale enterprise platforms, including a 643-class mortgage solution processing 1.7M requests/day with 99.99% success, alongside production processing of 2.8M+ pages from 49 encrypted lender sources.
- Deep Learning specialist with hands-on expertise across PyTorch, TensorFlow, Transformers, CNNs, ResNet50, BiLSTM, Swin/ViT, Donut, multimodal fusion, LoRA/QLoRA, PEFT, SFT, and RLHF/DPO, delivering models with up to 0.975 weighted F1 through distributed and multi-GPU training.
- Generative AI, LLM & RAG practitioner with knowledge and practical exposure to LLM-based applications, Retrieval-Augmented Generation (RAG), FAISS, BM25, dense retrieval, Reciprocal Rank Fusion (RRF), cross-encoder reranking, structured outputs, Llama 3, and Claude API, with growing focus on Agentic AI and modern GenAI architectures.
- End-to-end ML engineering professional experienced in building scalable ML workflow across model optimization, selective prediction, confidence calibration, MLOps, MLflow, AWS SageMaker, ONNX, Docker, Jenkins CI/CD, CloudWatch, Spark/PySpark, and Databricks, with a strong focus on taking ML solutions from experimentation to reliable production systems.
- Architecture-oriented ML engineer with a strong understanding of designing scalable ML pipelines and production AI systems, including data-to-model workflow, deployment, monitoring, reliability, incident RCA, cost optimization, and release governance, with the capability to contribute across the full ML lifecycle as a senior technical individual contributor.

TECHNICAL SKILLS

- Machine Learning & Deep Learning: PyTorch | PyTorch Lightning | TensorFlow | Keras | Hugging Face Transformers | CNN | ResNet50 | BiLSTM | Swin Transformer | Vision Transformer (ViT) | Donut | Multimodal Fusion | Named Entity Recognition (NER) | LoRA | QLoRA | PEFT | Supervised Fine-Tuning (SFT) | Instruction Tuning | RLHF/DPO | Mixed Precision (FP16/BF16)
- Generative AI, LLM & RAG: Large Language Models (LLMs) | Generative AI | Retrieval-Augmented Generation (RAG) | FAISS | Dense Retrieval | Okapi BM25 | Hybrid Retrieval | Reciprocal Rank Fusion (RRF) | Cross-Encoder Reranking | Structured Outputs | JSON Schema | Llama 3 | Anthropic Claude API | Selective Prediction | Confidence Calibration
- Computer Vision & Document AI: Computer Vision | OCR/OCR-Free Processing | OpenCV | Tesseract | AWS Textract | TensorFlow Object Detection | Image Preprocessing | Document Classification | Information Extraction | Object Detection | Signature Verification | Document Deduplication
- Distributed ML & Data Engineering: Distributed Data Parallel (DDP) | Horovod | TorchDistributor | NCCL | Multi-GPU Training | Apache Spark | PySpark | Databricks
- MLOps, Deployment & Model Serving: AWS SageMaker | MLflow | Model Registry | Real-Time Model Endpoints | Auto Scaling | TensorFlow Serving | ONNX | Docker | Jenkins CI/CD | AWS CloudWatch | Model Optimization | Production Monitoring
- Programming, Databases & Platforms: Python | SQL | Git | Linux | Bash

CAREER TIMELINE



CORE COMPETENCIES

- Document AI & Intelligent Document Processing
- Machine Learning & Multimodal Deep Learning
- Computer Vision & OCR-Free AI
- NLP & Information Extraction
- LLM Engineering & Generative AI
- RAG & Enterprise Retrieval Systems
- Transformer Architecture & Model Fine-Tuning
- Distributed ML & Large-Scale Model Training
- MLOps & Production AI Engineering
- AWS SageMaker & Cloud AI Engineering
- Model Optimization, Inference & AI Reliability
- AI Solution Architecture & Technical Leadership

EDUCATION

- B.Tech. – Computer Science & Engineering: TKR Engineering College, JNTU, Hyderabad | 2016

CERTIFICATIONS

- Please mention if any

PERSONAL DETAILS

- Date of Birth: DD/MM/YY
- Languages Known: English & Hindi

WORK EXPERIENCE

ICE DATA SERVICES | Senior ML Engineer | Hyderabad | Nov 2023 – Present

- Owned the end-to-end ML lifecycle for ICE's mortgage document classification platform spanning **643 document classes** and **2.8M+ training pages** from 49 encrypted lender sources, supporting **1.7M requests/day** and **45M+ requests/month** on AWS SageMaker with **1.8-second p95 latency** and **99.99% success rate**; reduced document verification turnaround from approximately **4 days to 4–5 hours**.
- Engineered a **five-branch multimodal classification architecture** combining **ResNet50, CNN, BiLSTM, and custom multi-head-attention Transformer components** with frozen GloVe embeddings; scaled distributed training across **20 A10G GPUs** using **Horovod**, resolving host-memory OOM and distributed all-reduce failures and achieving **0.959–0.972 weighted F1** across five lender benchmarks.
- Developed an **OCR-free Donut/Swin-based classification model** with **75M fully fine-tuned parameters**, training through PyTorch Lightning DDP across **20 L40S GPUs** using **BF16**; achieved **0.975 weighted F1 on 528K+ pages** versus **0.974 F1** from the OCR-based production ensemble, and exported the model to **ONNX** after benchmarking **9 SageMaker instance types**, delivering approximately **20% lower hourly infrastructure cost**.
- Improved model performance across **946K+ pages** by identifying a train-to-benchmark taxonomy mismatch that had reduced F1 from **0.62 to 0.92**; further established that **64% of residual errors were valid predictions discarded by confidence thresholds**, leading to calibrated **per-class selective prediction** across all 643 classes.
- Led technical RCA for a **P1 production regression** involving **21,096 reported failures across 260K pages**; traced five downstream services using **CloudWatch Logs Insights**, ruled out model and infrastructure outage hypotheses, and identified a **Java platform validation defect** that rejected complete documents when individual pages failed, establishing the actual failure rate at **2.3%**.
- Managed production ML operations and releases, delivering **four consecutive model releases** through the **Jenkins-to-SageMaker CI/CD pipeline**, including cross-account artifact promotion, model registration, staging validation, change control, rollback, and deployment; scaled production infrastructure from **20 to 118 SageMaker instances** during demand surges while maintaining reliability.
- Modernized the production serving stack by migrating model exports from **TensorFlow 1.x to TensorFlow 2.x**, improving maintainability and framework alignment while supporting reliable production inference.

PERSISTENT SYSTEMS | Senior ML Engineer | Client: ICE Data Services | Hyderabad | Aug 2021 – Nov 2023

- Engineered the foundational data ingestion and OCR pipeline supporting ICE's production document classification ecosystem, processing encrypted lender image sets, extracting text at scale with **Tesseract**, and consolidating **27+ source schemas** into a unified labelled corpus using **PySpark UDFs**.
- Developed an **OpenCV-based document preprocessing pipeline** incorporating thresholding, denoising, skew correction, and image normalization, improving OCR quality and downstream model performance across diverse lender inputs.
- Built the first multimodal **CNN + BiLSTM training pipeline** on Databricks, including the BiLSTM encoder-decoder responsible for generating sequence representations consumed by downstream text-processing branches.
- Established the initial **AWS SageMaker deployment and monitoring architecture**, together with **MLflow experiment tracking**, providing the foundation for the subsequent production MLOps and Jenkins-to-SageMaker release pipeline.
- Supported the evolution of the initial ML architecture into the **production-scale 643-class document classification ensemble**.

ZYCLYX | Data Scientist | Client: Al Rajhi Bank, Riyadh | Riyadh | Aug 2019 – Mar 2021

- Designed and built the bank's document digitization platform as its first ML Engineer, processing approximately **100K bilingual Arabic/English documents** through an end-to-end classification, detection, OCR, and entity-extraction pipeline.
- Developed a **ResNet50 document classifier** covering **~65 document types**, achieving approximately **88% classification accuracy**, followed by TensorFlow Object Detection for field localization, region cropping, Tesseract OCR, and GRU-based NER for extracting **names, countries, identification numbers, and other key entities**.
- Engineered a **deep-learning signature verification model** to identify potentially forged signatures and strengthen document-level fraud detection.
- Diagnosed the primary upstream source of OCR errors by auditing scanning configurations across major bank branches; identified inconsistent scanner settings causing colour distortion, rotation, and skew, and established a standardized configuration subsequently mandated bank-wide through an official bank circular.
- Integrated ML extraction services into the bank's **BluePrism RPA workflow** with a human-in-the-loop validation layer, enabling page-level and field-level verification while feeding corrections back as training signals.
- Productionized the platform on-premise using **Flask and Gunicorn microservices**, ensuring sensitive documents remained within the bank's network.
- Mentored **5+ engineers** as the ML team expanded, accelerating their capabilities across OCR, document processing, model development, and training workflow.

VISSINDIA | Python & Computer Vision Engineer | Hyderabad | Feb 2017 – Aug 2019

- Developed real-time computer-vision solutions for identity verification and multi-object tracking using **Haar Cascades, HOG, CNNs, SIFT, SURF, FLANN, TensorFlow Object Detection, and OpenCV**.
- Built image and video preprocessing pipelines to improve the quality and consistency of noisy visual inputs before model inference.
- Developed detection, feature-matching, and tracking workflow for real-time video analytics and identity-verification applications.

MAJOR PROJECTS

Rulebook-RAG | Retrieval-Grounded Document Classification | Personal Project | 2026

- Built a **36-class RAG classifier** without labelled training data or model retraining, using a dynamically indexed rulebook.
- Enabled **zero-retraining class expansion**, achieving **0.83 accuracy** on 5 newly added classes.
- Implemented **BM25 + dense retrieval + RRF**, improving accuracy by **5.2 pts**; cross-encoder reranking added another **3.9 pts**.
- Delivered **citation-grounded predictions** with live rule validation, achieving **0 invalid citations across 775 predictions**.

Document Deduplication Pipeline | ICE Data Services

- Built near-duplicate detection across a **7.1M-page mortgage corpus** using **Donut/Swin embeddings** across **40 GPU nodes**.
- Implemented **two-tier FAISS cosine search** at page and document levels using mean-pooled embeddings.
- Optimized exact retrieval through **label/page-count grouping** and transitive **Union-Find clustering**.
- Identified **67% of pages** and **69% of documents** as near-duplicates for large-scale corpus rationalization.